Dylan C. Gaeta

Curriculum Vitae

dylan.gaeta@colorado.edu
dylangaeta.com

325 Broadway
Boulder, CO 80305

Research Experience

Post-Doctoral Research Associate 03/2026 – Present
Cooperative Institute for Research in Environmental Sciences (CIRES),
University of Colorado Boulder
NOAA Global Monitoring Laboratory Boulder, CO

Doctoral Research, Greenhouse Gas Lab 01/2020 – 02/2026
Johns Hopkins University, Department of Environmental Health & Engineering Baltimore, MD

Applied Data and Governance Fellow 01/2019 – 01/2020
International Innovation Corps, The University of Chicago Chicago, IL

Graduate Research Assistant 08/2016 – 08/2018
University of Chicago, Department of the Geophysical Sciences Chicago, IL

International Student Research Experience (NSF iREU) 05/2015 – 08/2015
Technische Universität Graz, Institut für Chemische Technologie von Materialien Graz, Austria

McNair Scholars Research Program / NSF REU 05/2014 – 05/2015
University of Rochester, Department of Chemistry Rochester, NY

Education

Ph.D., Geography & Environmental Engineering 01/2020 – 02/2026
Johns Hopkins University Baltimore, MD

- *Dissertation*: "U.S. emissions of trace gas air pollutants inferred from a long-term atmospheric measurement network: from fumigants to fossil fuels" (Advisor: Prof. Scot M. Miller)

M.S., Geophysical Sciences 08/2016 – 08/2018
The University of Chicago Chicago, IL

- *M.S. Thesis*: "An Adjoint Trajectory Model of the Perturbed Carbon Cycle"

B.S., Applied Mathematics 08/2012 – 05/2016
B.A., Chemistry 08/2012 – 05/2016
University of Rochester Rochester, NY

- *Honors*: Graduated *cum laude* with Highest Distinction in Chemistry, with Distinction in Mathematics, McNair Scholar, Chemistry Department Award, Undergraduate Teaching Award, Junior Scholar Award, Dean's List 8 of 8 semesters, Chemistry Workshop Leader, NCAA Division III Track & Field Athlete

Publications

Peer Reviewed Journals:

- Z. Huang, A. Chen, L. Feng, W. S. Daniels, **D. C. Gaeta**, K. L. Morgan, L. T. Murray, X. Yu, B. F. Hobbs, S. M. Miller, "Satellite-based emissions estimate indicates progress toward China's methane mitigation goals." *Communications Earth & Environment* (In Review, 2026). https://arxiv.org/abs/2607.16931

- H. Liu, C. E. Miller, C. Sweeney, J. D. Watts, R. Commane, L. D. Schiferl, Z. Zhang, B. Poulter, L. Feng, A. Chen, Z. Huang, **D. C. Gaeta**, K. L. Morgan, W. S. Daniels, Q. Li, J. R. Melton, V. Yadav, D. Huntzinger, S. M. Miller, "Atmospheric observations reveal missing CH_4 fluxes from the North Slope of Alaska." *Atmospheric Chemistry and Physics* (In Review, 2026).
- **D. C. Gaeta**, J. Mühle, L. Hu, S. Montzka, I. J. Vimont, M. Crotwell, M. Zhang, Z. Huang, A. Chen, H. Liu, S. Milletich, D. Waugh, S. M. Miller, "A decline in terrestrial methyl bromide emissions from the U.S. since the Montreal Protocol." (In Prep, 2026).
- **D. C. Gaeta**, I. J. Vimont, M. Crotwell, L. Hu, K. McKain, B. C. Baier, Z. Huang, A. Chen, H. Liu, W. S. Daniels, L. T. Murray, P. DeCarlo, S. M. Miller, "Emissions of propane, butane, and pentane from fossil fuel activities in the U.S.." *In Preparation* (2026).
- A. Chen, W. S. Daniels, Z. Huang, A. P. Ravikumar, S. M. Jordaan, Z. Zhang, L. Feng, H. Liu, L. T. Murray, X. Yu, **D. C. Gaeta**, K. L. Morgan, J. Yuan, S. M. Miller, "U.S. oil and gas methane emission intensity declined 2019–2024 with little indication of federal policy impact." *Communications Earth & Environment* (2026). https://doi.org/10.1038/s43247-026-04039-0
- H. Liu, M. Ishizawa, F. R. Vogel, Z. Zhang, B. Poulter, L. Feng, A. Chen, A. L. Gagné-Landmann, D. N. Huntzinger, J. R. Melton, V. Yadav, **D. C. Gaeta**, Z. Huang, D. E. J. Worthy, D. Chan, S. M. Miller, "Methane fluxes from Arctic & boreal North America: comparisons between process-based estimates and atmospheric observations." *Atmospheric Chemistry and Physics* (2026). https://doi.org/10.5194/acp-26-1229-2026
- A. Chen, Z. Zhang, B. Poulter, L. Feng, H. Liu, L. T. Murray, X. Yu, Z. Huang, **D. C. Gaeta**, K. L. Morgan, Q. Zhu, S. M. Miller, "Spatial and temporal distribution of global wetland methane emissions during 2019–2020 estimated from satellite observations." *JGR Atmospheres* (2025). https://doi.org/10.1029/2025JD044354
- B. W. Clouser, L. C. Sarkozy, C. E. Singer, C. C. KleinStern, A. Desmoulin, **D. C. Gaeta**, S. Khaykin, S. Gabbard, S. Shertz, E. J. Moyer, "The Airborne Chicago Water Isotope Spectrometer: An Integrated Cavity Output Spectrometer for Measurements of the HDO/H_2O Isotopic Ratio in the Asian Summer Monsoon." *Atmospheric Measurement Techniques* (2025). https://doi.org/10.5194/amt-18-6465-2025
- **D. C. Gaeta**, J. Mühle, I. J. Vimont, M. Crotwell, L. Hu, J. B. Miller, K. McKain, B. C. Baier, M. Zhang, J. Bao, B. R. Miller, S. M. Miller, "California dominates U.S. emissions of the pesticide and potent greenhouse gas sulfuryl fluoride." *Communications Earth & Environment* (2024). https://doi.org/10.1038/s43247-024-01294-x
- M. Zhang, I. J. Vimont, S. M. Jordaan, L. Hu, K. McKain, M. Crotwell, **D. C. Gaeta**, S. M. Miller, "U.S. ethane emissions and trends estimated from atmospheric observations." *Environmental Science & Technology* (2024). https://doi.org/10.1021/acs.est.4c00380
- R. Lei, J. Poe, D. Huntzinger, J. Liu, S. Sitch, D. F. Baker, L. Feng, **D. C. Gaeta**, Z. Huang, S. M. Miller, "The Orbiting Carbon Observatory-2 (OCO-2) and in situ CO_2 data suggest a larger seasonal amplitude of the terrestrial carbon cycle compared to many dynamic global vegetation models." *Remote Sensing of Environment* (2024). https://doi.org/10.1016/j.rse.2024.114326
- L. Feng, S. Tavakkoli, S. M. Jordaan, A. E. Andrews, J. S. Benmergui, D. W. Waugh, M. Zhang, **D. C. Gaeta**, S. M. Miller, "Inter-annual Variability in Atmospheric Transport Complicates Estimation of US Methane Emissions Trends." *Geophysical Research Letters* (2023). https://doi.org/10.1029/2022GL100366

Other Works:

- **D. C. Gaeta**, J. Alumbro, K. Negandhi, C. Liang, D. Moon, S. M. Miller, "Greenhouse Gas Emissions Inventory for the City of Baltimore, 2018–2020" (2022). *Prepared for the Baltimore Office of Sustainability.* https://www.baltimoresustainability.org/wp-content/uploads/2023/04/2020_Baltimore_GHG_inventory_v2.pdf
- **D. C. Gaeta**, K. Negandhi, C. Liang, M. Kleckner, S. M. Miller, "City of Baltimore 2017 Greenhouse Gas Emissions Inventory Report" (2020). *Prepared for the Baltimore Office of Sustainability.* https://www.baltimoresustainability.org/wp-content/uploads/2021/09/2017_Baltimore_Inventory_v5-1.pdf
- **D. C. Gaeta**, "An Adjoint Trajectory Model of the Perturbed Carbon Cycle" (2018). *Knowledge@UChicago* (M.S. Thesis). https://doi.org/10.6082/uchicago.3551

Media Coverage

News Articles:

- "California the culprit for spike in little-known greenhouse gas more potent than CO_2" by Tom Perkins, *The Guardian* (April 2024). https://www.theguardian.com/us-news/2024/apr/19/california-toxic-gas-sulfuryl-fluoride
- "Thanks to termite tents, California is top U.S. emitter of a planet-warming pesticide" by Melody Petersen, *Los Angeles Times* (April 2024). https://www.latimes.com/environment/story/2024-04-03/california-is-biggest-us-emitter-of-this-greenhouse-gas
- "California is a hot spot for an untracked greenhouse gas that's way more potent than CO_2" by Jack Lee, *San Francisco Chronicle* (April 2024). https://www.sfchronicle.com/climate/article/greenhouse-gas-emission-california-19379362.php
- "California Leads the Nation in Emissions of a Climate Super-Pollutant, Study Finds" by Phil McKenna & Liza Gross, *Inside Climate News* (April 2024). https://insideclimatenews.org/news/03042024/california-sulfuryl-fluoride-emissions/
- "A Rare Greenhouse Gas Comes from—Termite Pesticide?" by Chelsea Harvey, *Scientific American* (April 2024). https://www.scientificamerican.com/article/a-rare-greenhouse-gas-comes-from-termite-pesticide/
- "California emits more of this little-known greenhouse gas than all other US states combined" by Sharon Udasin, *The Hill* (April 2024). https://thehill.com/policy/equilibrium-sustainability/4570418-california-greenhouse-gas-termites/
- "California Revealed As Worst US Emitter of 'Under the Radar' Greenhouse Gas" by Hillel Aron, *Courthouse News Service* (April 2024). https://www.courthousenews.com/under-the-radar-greenhouse-gas-seen-mostly-in-southern-california/
- "California Leads U.S. Emissions of Little-Known Greenhouse Gas" by Hannah Robbins, *Johns Hopkins Communications* (April 2024). https://hub.jhu.edu/2024/04/03/california-leads-us-emissions-of-little-known-greenhouse-gas/
- "Researcher sheds light on the main source of a rare but destructive greenhouse gas" by Danielle Underferth, *Johns Hopkins Communications* (Mar. 11, 2022). https://hub.jhu.edu/2022/03/11/sulfuryl-fluoride-greenhouse-emissions/
- "Termite Fumigation in California Is Fueling the Rise of a Rare Greenhouse Gas" by Jenessa Duncombe, *Eos Science News by AGU* (Jan. 3, 2022). https://eos.org/articles/termite-fumigation-in-california-is-fueling-the-rise-of-a-rare-greenhouse-gas

Conference Presentations

CIRES Rendezvous

- **D. C. Gaeta**, J. Mühle, L. Hu, I. J. Vimont, M. Crotwell, S. Montzka, S. M. Miller, "A decline in U.S. methyl bromide emissions throughout 2007–2021." *Environmental Monitoring and Observations* (Poster). Boulder, CO (2026).

American Geophysical Union Fall Meeting

- **D. C. Gaeta**, M. Zhang, L. Feng, I. J. Vimont, M. Crotwell, L. Hu, K. McKain, B. C. Baier, J. B. Miller, A. Andrews, S. M. Miller, "Emissions of C_3–C_5 alkanes from the U.S. fossil fuel industry and implications for air quality." *Emissions and Air Quality Impacts of Atmospheric Pollutants from Oil, Gas, and Coal Operations* (Oral Session A42E-06). Washington, D.C. (2024).
- **D. C. Gaeta**, J. Mühle, L. Hu, I. J. Vimont, S. Montzka, M. Crotwell, M. Zhang, K. McKain, B. C. Baier, J. B. Miller, S. M. Miller, "U.S. emissions of methyl bromide have declined since the Montreal Protocol phase-out, but not to zero." *General Atmospheric Chemistry, Urban Winter Air Quality, and Constraints on Atmospheric Oxidation Capacity* (Poster Session A21H-1855). Washington, D.C. (2024).
- **D. C. Gaeta**, J. Mühle, L. Hu, I. J. Vimont, S. Montzka, M. Crotwell, K. McKain, B. C. Baier, J. B. Miller, M. Zhang, L. Feng, A. Chen, H. Liu, S. M. Miller, "Trends in California's methyl bromide emissions from 2004–2022 estimated using atmospheric measurements and inverse modeling." *Scientific Impact of Long-Term Observations of the Atmosphere* (Oral Session A24I-08). San Francisco, CA (2023).
- **D. C. Gaeta**, J. Mühle, I. J. Vimont, M. Crotwell, K. McKain, B. C. Baier, J. B. Miller, M. Zhang, J. Bao, B. R. Miller, S. M. Miller, "Constraining U.S. Sulfuryl Fluoride Emissions with Atmospheric Measurements and a Geostatistical Inverse Model." *Stratospheric and Tropospheric Composition Changes: Observations and Modeling of Special Events, Feedback Mechanisms, and Long-Term Trends* (Poster Session A52Q-1204). Chicago, IL (2022).
- **D. C. Gaeta**, J. Mühle, I. J. Vimont, M. Zhang, K. McKain, M. Crotwell, B. R. Miller, S. M. Miller, "Greater Los Angeles Area dominates U.S. emissions of sulfuryl fluoride, a potent greenhouse gas." *Constraining Greenhouse Gas Exchange Processes Using Remote Sensing and in Situ Observations* (Poster Session A15K-1789). New Orleans, LA (2021).
- **D. C. Gaeta**, I. J. Vimont, B. R. Miller, S. M. Miller, "California dominates U.S. emissions of sulfuryl fluoride, a potent greenhouse gas." *Constraining Greenhouse Gas Exchange Processes Using Remote Sensing and in Situ Observations* (Lightning Talk/iPoster A128-11). Virtual (2020).
- **D. C. Gaeta**, D. R. MacAyeal, "An Adjoint Trajectory Model of the Perturbed Carbon Cycle." *Coupled Human–Natural Systems and Global Environmental Change: Innovative Interdisciplinary Approaches* (Poster Session GC31D-1291). Washington, D.C. (2018).

Teaching Experience

Teaching Assistant / Field Trip Co-leader 2021 – 2025
Johns Hopkins University, Department of Environmental Health & Engineering Baltimore, MD

- Case Studies in Climate Change: California (Seminar) (EN.570.320) (Prof. S. Miller)
- Case Studies in Climate Change (Field Course) (EN.570.321) (Prof. S. Miller)
    - Co-led an annual 8-day field course for 10 undergraduate students in Death Valley National Park and the Eastern Sierras that focuses on past, present, and future climate change in California.

SOUL Instructor (Special Opportunities for Undergraduate Learning) Fall 2023
Johns Hopkins University Baltimore, MD

- SOUL: Greenhouse Gases in Earth's Atmosphere (AS.360.111)
    - Wrote a proposal and was successfully funded to teach a new 1.0-credit seminar course for 10 students that explored the mechanistic (process-based) relationship between GHGs and Earth's climate.

Teaching Assistant 2016 – 2018
University of Chicago, Physical Sciences Division and The College Chicago, IL

- Global Warming: Understanding the Forecast (GEOS 23400) (Prof. D. Archer)
- Energy: Science, Technology, and Human Usage (GEOS 24750) (Prof. E. Moyer)
- Natural Hazards (PHSC 13600) (Prof. N. Nakamura)
- Earth as a Planet: Exploring Our Place in the Universe (PHSC 10800) (Prof. F. Ciesla)

Workshop Leader / Teaching Assistant 2013 – 2016
University of Rochester, Department of Chemistry Rochester, NY

- Molecular Spectroscopy (CHM 232) (Laboratory manual revisions) (Prof. L. Rothberg)
- Physical Chemistry II: Molecular Thermodynamics (CHM 252) (Prof. J. Farrar)
- Physical Chemistry I: Quantum Chemistry (CHM 251) (Prof. L. Rothberg)
- Organic Chemistry I & II (CHM 203 & 204) (Prof. J. Goodman and Prof. B. Nilsson)
- Chemical Concepts, Systems, and Practice I & II (CHM 131 & 132) (Prof. J. Farrar and Dr. B. Hafensteiner)

Leadership and Outreach

EHESO Sustainability Leadership Council Representative 2023 – 2025
Environmental Health & Engineering Student Organization, Johns Hopkins University Baltimore, MD

- https://sustainability.jhu.edu/who-we-are/slc/
- Elected to serve as the Department representative on the university-wide Sustainability Leadership Council. Served as a grant reviewer for the Campus as a Living Lab grant funding program: https://sustainability.jhu.edu/research/livinglab/

EHESO Graduate Representative Organization (GRO) Representative 2023 – 2024
Environmental Health & Engineering Student Organization, Johns Hopkins University Baltimore, MD

- https://studentaffairs.jhu.edu/gro
- Elected to serve as the Environmental Health & Engineering Department Representative for the Graduate Representative Organization for Student Affairs on the Homewood campus.

Undergraduate Research Mentor 2021 – 2025
Greenhouse Gas Lab, Johns Hopkins University Baltimore, MD

- https://greenhousegaslab.org/
- Mentored and collaborated with Krittika Negandhi, Cheryl Liang, Mitchell Kleckner, Julia Alumbro, Daniel Moon, and Millie Mi on undergraduate research projects focused on urban emissions in Baltimore and other U.S. cities.

WISE High School Student Research Mentor 2022 – 2023
Johns Hopkins University, Center for Educational Outreach, Western High School Baltimore, MD

- https://engineering.jhu.edu/outreach/ceo-programs/wise/
- Mentored and advised two Baltimore City high school students, Oumou Sall and Saria Malik, throughout paid internships at JHU. Designed and supervised projects to develop technical skills while focusing on greenhouse gas emissions from cities.

Johns Hopkins Science Policy Group Member 2020 – 2025
Johns Hopkins University, Homewood Campus Baltimore, MD

Young Alumni Council (Chicago Network) 2018 – 2023
University of Rochester Chicago, IL

Co-organizer, Climate and Energy Student Lunch & Learn weekly series 2017 – 2018
University of Chicago – EPIC/RDCEP Chicago, IL

Honors and Awards

Outstanding Doctoral TA Award 2024
Johns Hopkins University, Department of Environmental Health & Engineering

Geyh-Bouwer Trainee Practice Award 2023
Johns Hopkins University, Department of Environmental Health & Engineering

- https://publichealth.jhu.edu/departments/environmental-health-and-engineering/research-and-practice/geyh-bouwer-trainee-practice-award/
- The Geyh-Bouwer Trainee Practice Award recognizes students whose application of science demonstrates the betterment of the environment and/or public health.

EHE Department Retreat Poster Award 2023
Johns Hopkins University, Department of Environmental Health & Engineering

GRO Conference Travel Grant 2020
Johns Hopkins University

Carl A. Whiteman Jr. Teaching Award 2016
University of Rochester

Ronald E. McNair Scholar 2013 – 2016
United States Department of Education

Chemistry Department Award 2016
University of Rochester

Junior Scholar Award 2015
University of Rochester, Department of Chemistry

Dean's Scholarship 2012 – 2016
University of Rochester

New York State Scholarship for Academic Excellence 2012 – 2016
NYS Higher Education Services Corporation

Chautauqua Region Community Foundation Scholarship 2012 – 2016
Chautauqua Region Community Foundation, Western New York